

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	
Project Title	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an AI-powered crop recommendation system that predicts the most suitable crop based on soil nutrients and environmental conditions, helping farmers make informed agricultural decisions.
Scope	The project analyzes soil nutrient values (N, P, K), temperature, humidity, pH, and rainfall to recommend the best crop using Machine Learning through a user-friendly web application.
Problem Statement	
Description	Farmers often face difficulties in selecting the most suitable crop due to varying soil conditions, changing weather patterns, and limited access to data-driven decision-making tools.
Impact	Providing accurate crop recommendations can improve agricultural productivity, reduce crop failure, optimize resource utilization, and support sustainable farming.
Proposed Solution	
Approach	Develop a Machine Learning-based crop recommendation system using soil and weather parameters to predict the most suitable crop through a Flask-based web application.
Key Features	AI-powered crop prediction Soil nutrient analysis Weather parameter analysis

	Fast and accurate recommendations Simple and user-friendly web interface
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 or higher
Memory	RAM	8 GB
Storage	SSD	256 GB or above
Software		
Frameworks	Web frameworks	Flask
Libraries	ML libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE	VS Code, Jupyter Notebook
Data		
Dataset	Crop Recommendation Dataset	CSV format
Source	Kaggle	Agricultural Crop Recommendation Dataset